Modelisation & Simulation Report:

Cellular Automata

Effect of curcumin on breast cancer developement

M1 GENIOMHE-AI

2025 -2026

Professor: Franck DELAPLACE

Name	Surname	Student Number
Boomika	SELVARADJOU	20220362
Sitty Nouzlat	SAID CHANFI	20235430
Lydia	SRINIVASSAN	20223604

Table of contents:

1. Introduction.....4
2. Analysis.....5
3. Model.....7
    a. Cellular automata framework.....7
    b. Biological states.....8
    c. Transition rules.....8
    d. Main model parameters.....9
4. Simulation scenarios.....11
5. Conclusion.....15
6. References.....16
7. Annexes.....18

Abbreviation:

- BC: Breast Cancer
- CCK-8: Cell Counting Kit-8
- EGFR: Epidermal Growth Factor Receptor
- ER: Estrogen Receptor
- Fer-1: Ferrostatin-1
- HER2: Human Epidermal growth factor Receptor 2
- HDAC: Histone Deacetylase
- IV: Intravenous (medical route of administration)
- miRNA/ miRNAs: microRNAs
- MDA: Malondialdehyde
- p21: Cyclin-Dependent Kinase Inhibitor 1A (CDKN1A)
- PI3K/Akt signaling pathway: Phosphoinositide 3-kinase/ Protein kinase B pathway
- PR: Progesterone Receptor
- ROS: Reactive Oxygen Species
- TNBC: Triple Negative Breast Cancer
- TSGs: Tumor Suppressor Genes
- Wnt/β-catenin signaling pathway: Wingless/ Integrated and Beta-catenin signaling pathway

Keywords:

- Breast Cancer
- Curcumin
- Cellular Automata

1. Introduction

Breast cancer (BC) is one of the most common cancers for women in the world, representing 30% of cancers and 14% of cancers-related deaths in women (Cao et al., 2022). Its development is heterogeneous and results from uncontrolled cell proliferation, cell-to-cell contact of tumor and other cells, signaling molecules, and the surrounding tissue microenvironment. Breast cancers are classified according to the presence of protein markers, into sub-types: ERα (estrogen receptor), PR (progesterone receptor) and ERBB2 (or HER2, human epidermal growth factor receptor 2). TNBC (triple negative breast cancer) is characterized by the absence of ERα, PR and HER2 markers. More than 70% of BC are ER+ (estrogen receptor), which can be treated with targeted hormone therapies because they express specific receptors. HER2 types can also be treated by monoclonal antibody treatment. In contrast, TNBC remains difficult to treat due to the lack of specific molecular targets, and is considered the most aggressive subtype, presenting clinical challenges (Mayo et al., 2024; Cao et al., 2022).

To overcome these limitations, recent studies have highlighted the therapeutic potential of curcumin, a natural polyphenol extracted from the rhizome of turmeric (Curcuma longa). Curcumin is well-known for its anti-inflammatory and anti-canceral properties, and has shown inhibitory effects on the survival of cancer cells. It targets epigenetic machinery (DNA methylation) (Fabianowska-Majewska et al., 2021), induces ferroptosis (Cao et al., 2022) and apoptosis in order to inhibit tumor growth. In addition, curcumin has shown to be effective in ER+ and TNBC breast cancers types. It can be used as an adjuvant therapy to reduce side-effects of conventional cancer treatments (ex: radiation dermatitis, Mayo et al., 2024) and by combining curcumin with other chemotherapeutic agents, it can enhance the drug absorption and therapeutic efficacy.

In our project, we aim to model the effects of curcumin on breast cancer cell development using cellular automata. Cellular automata are computational models based on local interactions between cells evolving over time according to predefined rules. The models are useful for simulating complex biological systems such as tumor growth and treatment response. Through this approach, we want to better understand the dynamics of cancer progression and the potential impact of curcumin on tumor growth.

2. Analysis

To model the effects of curcumin on breast cancer development using cellular automata, we first examined the biological mechanisms described in recent studies on curcumin and breast cancer treatment.

In the article “Mayo et al., Curcumin Administration Routes in Breast Cancer Treatment (2024)”, curcumin has shown anti-tumoral and antiproliferative effects in both in vivo and in vitro studies. Curcumin was shown to reduce cancer cell proliferation and induce apoptosis, particularly when combined with other chemotherapeutic agents such as docetaxel, paclitaxel, or doxorubicin [1, 2] (Annex: Table a & b). Clinical trials also reported improved cytotoxicity and reduced cancer cell viability when curcumin was administered using nanoemulsions or keratin nanoparticles [3], which enhances drug absorption and bioavailability.

Several molecular pathways involved in cancer progression are affected by curcumin, such as PI3K/Akt, EGFR, and Wnt/β-catenin, leading to reduced proliferation and increased autophagy and apoptosis. Curcumin also regulates gene expression by increasing tumor suppressor miRNAs while decreasing oncogenic miRNAs [4]. Also, curcumin inhibits NF-kB activation, by preventing the transcription of oncogenic genes and promotes the production of ROS (reactive oxygen species), leading to cancer cell death.

However, curcumin presents some pharmacokinetic limitations. Due to its hydrophobic nature, it is poorly absorbed when administered orally and is rapidly metabolized by the liver, resulting in a very low bio-availability. Although this mode of administration is inexpensive and non-invasive, intravenous administration provides higher bio-availability because it bypasses the intestinal metabolism. However, IV delivery is invasive, more costly and invasive, and reduces tolerability.

Therefore, nanoparticles and liposome based delivery systems seem to be the better options, by ensuring curcumin stability, better absorption and increased therapeutic efficiency [3].

Based on these observations, we defined some cellular automata parameters for the model to simulate the influence of different curcumin administration modes on the tumor evolution:

Parameter	Value	Justification
γ_{admin}	Oral = 0.20 / IV = 0.70 / Nano = 0.85	Encodes bioavailability differences across delivery routes, consistent with curcumin pharmacokinetic variability (Mayo et al., 2024).
k_{decay}	0.023	Exponential decay of effective curcumin activity over time, reflecting decreasing systemic efficacy after administration (Mayo et al., 2024).
D_0	1.0	Normalized initial drug concentration (dimensionless scaling factor for CA field).
T_{treat}	6 (standard) / 10 (nano)	Effective exposure window required to trigger downstream apoptotic/ferroptotic cascades. Matches reported delayed biological response times (4-6 cell cycles typical).

We then tried to understand the role of curcumin-induced ferroptosis in breast cancer, using the study "Cao X et al. Curcumin suppresses tumorigenesis by ferroptosis in breast cancer (2022)". In this study, the effects of curcumin were analyzed using breast cancer cell lines (MDA-MB-453 and MCF-7), using Cell Counting Kit-8 (CCK-8) assays. Demonstrating that curcumin suppresses cancer cell viability in a dose-dependent manner (Annex: Fig a), with significant reductions observed at concentrations between 15 to 30 μM after 24-72 hours of treatment (Annex: Fig b).

The study showed that curcumin induces ferroptosis, an iron-dependent form of programmed cell-death, different from apoptosis. Ferroptosis is characterized by lipid peroxidation caused by iron-catalyzed oxidation of unsaturated fatty acids within the cell membrane. Curcumin increased intracellular ROS, lipid peroxidation, MDA (malondialdehyde) accumulation and intracellular Fe2+ levels, which lead to ferroptosis. Also, curcumin upregulates the expression of SLC1A5 mRNA (Annex: Fig c), enhancing glutamine uptake and promoting metabolic pathways associated with ferroptosis [5].

We then defined some more cellular automata parameters to implement into the model, based on the effect of curcumin on ferroptosis:

Parameter	Value	Justification
$\gamma_{cyto}(D)$	0.05 → 0.40	Dose-dependent cytotoxic probability reflecting ferroptosis-driven viability reduction (Cao et al., 2022; 25-30 μM range).
$\tau_{ferroptosis}$	1-3 steps	Faster death cascade due to lipid ROS accumulation and GPX4 inhibition (Cao et al., 2022).

Finally, we studied the effect of curcumin on DNA methylation in breast cancer based on the study "Fabianowska-Majewska et al. Curcumin from Turmeric Rhizome: A Potential Modulator of DNA Methylation Machinery in Breast Cancer Inhibition. (2021)". The study highlights the effect of curcumin in epigenetic regulation, such as DNA regulation. In breast cancers, hypermethylation of tumor suppressor genes (TSGs) can silence protective cellular mechanisms, while hypomethylation will activate oncogenes and pro-metastatic genes [6]. DNA methylation plays a key role in the regulation of cell proliferation, DNA repair, cell differentiation, intracellular signaling and apoptosis.

Curcumin targets several signaling pathways associated with breast cancer development, such as Wnt/β-catenin, Notch and Hedgehog [7]. It also influences epigenetic regulation by regulating DNA methylation, by inhibiting HDAC (histone deacetylases) [8] and by regulating the expression of oncogenic and tumor suppressive miRNAs. This contributes to cell-cycle arrest through p21 upregulation [9], activation of 9-caspace and induction of apoptosis in cancer cells.

Curcumin uptakes are higher in malignant breast cancer cells than in normal cells, due to the differences in intracellular lipid composition and the increased presence of cytoplasmic lipid droplets in cancer cells.

Based on this, we define some parameters inspired by the effects of curcumin on DNA methylation and apoptosis:

Parameter	Value	Justification
$\tau_{apoptosis}$	4-6 steps	Time delay between "Cancerous → Treated → Dead" under epigenetic/apoptotic pathway (Fabianowska-Majewska et al., 2021).
S_{eff} (effective suppression strength)	0.17-0.35 (depends on γ_admin)	Aggregated effect of curcumin on apoptosis induction + epigenetic modulation + tumor suppression. Derived by compressing CURCUMIN_EFFECT (0.295) and CURC_TUM_RED (0.58) into a single bounded probability influence.

3. Model

a. Cellular automata framework

A Cellular Automaton (CA) is formally defined as a tuple < d, Q, N, f > where:

- d : dimension of the space (here d = 2, a 2D grid representing a breast tissue section)
- Q : finite set of cell states (biological compartments)
- $N \subseteq \mathbb{Z}^d$: neighborhood specification (Moore or Von Neumann)
- $f : Q^{|N|} \rightarrow Q$: local transition function applied synchronously and isotropically to all cells

The model uses a 2D Moore neighbourhood (8 neighbouring cells, with radius 1), which allows isotropic cell-cell interaction within the tissue (no privileged direction exists in the neighborhood for determining the evolution).

The grid uses torus topology (periodic boundary conditions) to eliminate artificial edge effects and simulate a continuous patch of breast epithelial tissue.

Cellular automata are particularly suitable for tumor modeling (Table 1) since global tumor behaviour emerges from local interactions between neighbouring cells, reproducing proliferation, invasion and tissue regeneration dynamics.

Table 1: Key properties of CAs ideal for biological tumor modeling (Delaplace, Paris-Saclay)

Property	Biological meaning
Collective	Tumor behavior emerges at the population (tissue) scale
Distributed	Each cell makes local decisions, no central tumor controller
Emerging	Global invasion patterns arise from simple local cell-cell interaction rules
Self-organized	The tumor microenvironment shapes itself through local interactions
Elementary	Each individual cell follows simple, biologically grounded transition rules

b. Biological states

We defined that Q is the following set of cell states {Healthy, Cancerous, Treated, Dead, Empty}. So, each grid position can occupy one of the following biological states:

State	Color	Biological meaning
Healthy (H)	Green	Normal healthy epithelial breast cell, susceptible to malignant transformation via oncogenic contact
Cancerous (C)	Red	Actively proliferating tumor cell, inducing malignant transformation in neighbouring healthy cells
Treated (T)	Yellow	Cancer cell treated by curcumin undergoing timed ferroptic or apoptotic cell death
Dead (D)	Grey	Eliminated cell post-ferroptosis or post-apotosis, biologically inert
Empty (E)	White	Vacant grid position after cell death/removal, extracellular space or cleared tissue

Each cell is encoded as a tuple: (state, τ). Where τ represents the internal timer associated with delayed biological processes such as apoptosis or regeneration.

c. Transition rules

We define the following transition rules defined between each states:

- ❖ Healthy(H) → Cancerous(C): Cancer propagation depends on the spontaneous mutation and local tumor influence through cell-cell contact.

We define: $P(H \rightarrow C) = \rho + \beta N_C$, where:

- ρ is the baseline malignancy probability,
- β represents neighborhood-driven tumor propagation,
- N_C is the number of cancerous neighbors.

- ❖ Cancerous (C) → Treated (T): Curcumin treatment depends on drug concentration and mode of administration.

We define: $P(C \rightarrow T) = 1 - e^{-S \cdot D(t)}$, where:

- S is the effective curcumin suppression strength,
- D(t) is the drug concentration field.

We define the exponential decrease of drug concentration over time: $D(t) = D_0 e^{-k_{decay} t}$.

- ❖ Cancerous (C) → Dead (D): direct cytotoxicity induced by ferroptosis.

We define: $P(C \rightarrow D) = \gamma_{cyto}(D)$, with increasing death probability for higher curcumin concentrations.

- ❖ Treated (T) → Dead (D): Treated cells undergoing apoptosis or ferroptosis after a delayed time.

We define: $T \rightarrow D$ after a time τ where:

- $\tau_{apoptosis}$= 4-6 steps,
- $\tau_{ferroptosis}$= 1-3 steps.

- ❖ Dead (D) → Empty (E): Dead cells are removed after some time.

We define: $D \rightarrow E = 1$.

- ❖ Empty (E) → Healthy (H): Tissue regeneration occurs locally when sufficient healthy neighbours are present.

We define: $P(E \rightarrow H) = \gamma_{reg} 1(N_H \geq N_H^{min})$ after τ_{reg} , 0 otherwise.

Here is the full transition table of our model:

Transition	Governing parameters
Healthy → Cancerous	ρ, β
Cancerous → Treated	S_{eff}, γ_{admin}, D(t)
Cancerous → Dead	γ_{cyto}(D)
Treated → Dead	$\tau_{apoptosis}$ or $\tau_{ferroptosis}$
Dead → Empty	deterministic removal (probability = 1)
Empty → Healthy	γ_{reg}

d. Main model parameters

Parameter	Value	Justification
γ_{admin}	Oral = 0.20 / IV = 0.70 / Nano = 0.85	Encodes bioavailability differences across delivery routes, consistent with curcumin pharmacokinetic variability (Mayo et al., 2024).
k	k_oral = 0.10 / k_fall_IV = 0.02/ k_fall_NANO = 0.005 / k_rise_IV = 0.3 / k_rise_NANO = 0.15	Exponential decay of effective curcumin activity over time, reflecting decreasing systemic efficacy after administration (Mayo et al., 2024), or rise in some cases.
D_0	ORAL = 0.2 NANO = 0.85	Normalized initial drug concentration (dimensionless scaling factor for CA field).

	IV = 0.95	
T_{treat}	6 (standard) / 10 (nano)	Effective exposure window required to trigger downstream apoptotic/ferroptotic cascades. Matches reported delayed biological response times (4–6 cell cycles typical).
S_{eff} (effective suppression strength)	0.05-0.2125 (depends on γ_{admin})	Aggregated effect of curcumin on apoptosis induction + epigenetic modulation + tumor suppression. Derived by compressing CURCUMIN_EFFECT (0.295) and CURC_TUM_RED (0.58) into a single bounded probability influence.
$\alpha_{apoptosis}$	implicit in S_{eff}	Represents HDAC inhibition, p21 activation, and pro-apoptotic signaling (Fabianowska-Majewska et al., 2021). Not modeled separately to avoid redundancy.
$\gamma_{cyto}(D)$	0.05 → 0.40	Dose-dependent cytotoxic probability reflecting ferroptosis-driven viability reduction (Cao et al., 2022; 25–30 μM range).
ρ	0.02 (standard) / 0.01 (TNBC)	Baseline spontaneous malignancy or immune escape probability (background tumor emergence noise, per experimental control conditions).
β	0.10 (standard) / 0.15 (TNBC)	Neighborhood-driven cancer propagation strength (Moore neighborhood influence, 8-cell adjacency).
N_{Moore}	8 neighbors	Standard Moore neighborhood defining isotropic local tissue interaction.
$T_{apoptosis} = T_treat$	4-6 steps	Time delay between "Cancerous → Treated → Dead" under epigenetic/apoptotic pathway (Fabianowska-Majewska et al., 2021). Not implemented.
$T_{ferroptosis} = T_treat$	1-3 steps	Faster death cascade due to lipid ROS accumulation and GPX4 inhibition (Cao et al., 2022). Not implemented.
T_{death}	deterministic after τ	Treated cells do not die instantly; they follow a fixed internal countdown before transitioning to Dead.
γ_{reg}	0.02-0.08	Probability that an Empty cell becomes Healthy when surrounded by viable tissue; models epithelial regeneration and wound-healing-like repopulation in breast tissue.
N_H^{min}	≥ 3 healthy neighbors	Minimum number of Healthy cells in Moore neighborhood required to trigger regeneration signal (local tissue integrity threshold).
τ_{reg}	2-5 steps	Delay before Empty → Healthy conversion, representing biological regeneration time (cell migration + division + adhesion restoration). Not implemented: direct probabilistic conversion made.

EPI	0.45 (default) / 0.2 (TNBC) / 0 (Ferro-only) / 0 (No treatment)	Controls epigenetic pathway activation.
$\gamma_{ferro}(D)$	min(0.30, 0.03 + 0.27 * D)	Probability of ferroptosis induction increases with drug activity.
$\gamma_{epi}(D)$	min(0.20, 0.02 + 0.18 * D)	Probability of epigenetic-mediated apoptosis increases with drug activity.

This cellular automata model integrates pharmacokinetics, ferroptosis, apoptosis, and epigenetic regulation into a simulation of breast cancer development.

By combining local cell interactions with curcumin-dependent transitions, the model allows the study of tumor growth dynamics under different curcumin administration strategies.

4. Simulation scenarios

We made some experimental scenarios to test different conditions of cancer subtypes, cellular mechanisms and mode of administration of curcumin. The administration route (Oral, IV, Nano) is selected at the launch via a pop-up window, determining the pharmacokinetic profile D(t). The biological scenarios are set directly in the code via boolean flags.
We implemented 4 scenarios, and note that the model distinguishes 2 independent dimensions: the biological mechanism (Scenarios 1, 2, 4) and cancer subtype (Scenario 3).

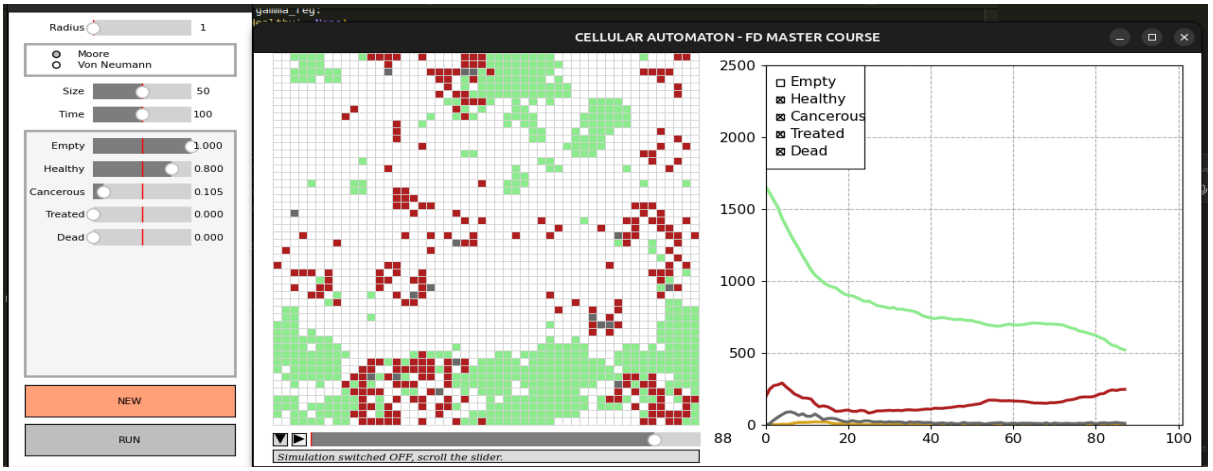
- Scenario 1: Ferroptosis pathway only (`SCENARIO_FERRO_ONLY = True`)

Biological question: If curcumin acts only through ferroptosis induction via SLC1A5 upregulation, can it inhibit tumour growth without epigenetic support ?

Biological justification: In the study by "Cao et al., 2022", curcumin was shown to dose-dependently suppress cell viability in breast cancer cell lines (MCF-7, MDA-MB-453), with an IC50 of approximately 20 µM. The cytotoxic effect was completely abolished by ferrostatin-1 (Fer-1), confirming that ferroptosis was a sufficient mechanism of cell death.
Through SLC1A5 upregulation, curcumin induces ferroptosis by increasing glutamine uptake, accumulation of lipid ROS, and membrane peroxidation leading to direct cell death. This scenario replicates the experimental condition where DNMT activity is preserved and only the ferroptotic pathway is active, equivalent to the Fer-1-free condition.

Parameters: gamma_admin = 0.7 (moderate drug delivery) ; EPI = 0 (epigenetic pathway disabled) ; T_treat = 6 (timer not used because ferroptosis kills directly) ; Curcumin_present = True (curcumin present) ; Beta = 0.1 (standard proliferation rate) ; Rho = 0.02 (standard immune clearance).

Expected simulation dynamics: Cancerous cells (red) die directly, transitioning to dead cells (grey) without going through a treated state (yellow). The dead cell grid expands quickly at the tumor core. Some cancerous cells persist at the periphery where the drug concentration D(t) is insufficient to cross the stochastic threshold at every step. We note that no yellow cells are visible since the epigenetic pathway has been disabled.



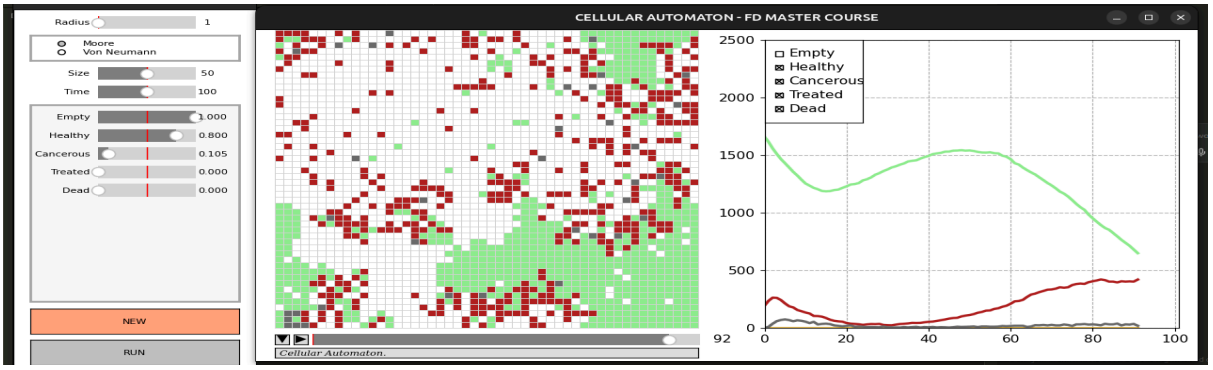
❖ Scenario 2: Epigenetic pathway only (`SCENARIO_EPI_ONLY = True`)

Biological question: If curcumin acts only through DNMT1 inhibition and tumour suppressor reactivation, can it suppress tumour proliferation without ferroptosis ?

Biological justification: In the study by "Fabianowska-Majewska et al., 2021", curcumin reduces DNMT1 mRNA and protein levels (by 2-4 folds), decreasing global DNA methylation (30-35%), and reactivating TSGs genes (RASSF1A, BRCA1, p21/CDKN1A). This leads to G1 cell cycle arrest and eventual apoptosis. The process is slower than ferroptosis, because it needs gene re-expression over many cell cycles before cell death.
We replicate the in vitro condition where ferroptosis is blocked by Fer-1 inhibitors.

Parameters: gamma_admin = 0 (sets S_{eff}=0, disabling ferroptosis) ; EPI = 0.45 (full epigenetic pathway enabled) ; T_treat = 6 (timer for steps between DNMT1 inhibition and cell death) ; Curcumin_present = True (curcumin present) ; Beta = 0.1 (standard proliferation rate) ; Rho = 0.02 (standard immune clearance).

Expected simulation dynamics: Cancerous cells (red) first transition to treated state (yellow) as DNMT1 inhibition triggers cycle arrest, then they progress to dead state (gray) after T_treat steps. A yellow ring should form at the tumor boundary and advances inwards over time. The progression is slower than ferroptosis (Scenario 1). The cancerous count declines gradually, and the comparison with ferroptosis illustrates the difference between the 2 different mechanisms at cell-inducing death.



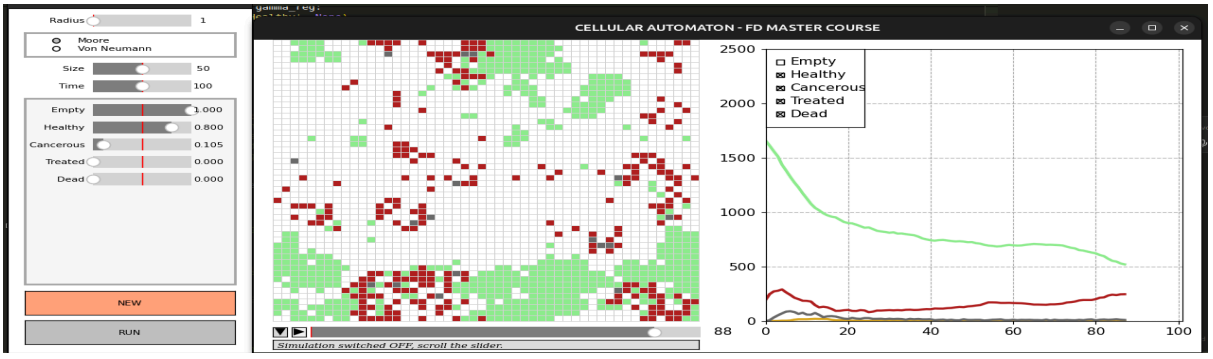
❖ Scenario 3: Triple-Negative Breast Cancer (TNBC) (`SCENARIO_TNBC = True`)

Biological question: How does curcumin perform against TNBC, with stronger immune evasion, faster proliferation, and reduced epigenetic sensitivity, compared to ER+ subtypes ?

Biological justification: Unlike scenarios 1 & 2, which isolate a specific cellular mechanism, this scenario models a specific cancer subtype, specifically the subtype TNBC, which is the most aggressive one (Mayo et al., 2024). This type is characterised by the activation of Hedgehog/Gli1 and PI3K/Akt signalling pathways driving faster proliferation, and stronger immune checkpoint dysregulation enabling immune evasion. We note that DNMT inhibition is less effective in TNBC because the epigenome is dysregulated, reducing the therapeutic impact of TSGs reactivation (Fabianowska-Majewska et al., 2021).
In this scenario, we replicate a TNBC patient receiving oral administration of curcumin.

Parameters: gamma_admin = 0.2 (lower drug delivery due to oral administration low bio-availability) ; EPI = 0.2 (reduced value of epigenetic pathway because TNBC epigenome is less responsive to DNMT inhibition) ; T_treat = 8 (longer steps to overcome cellular resistance) ; Curcumin_present = True (curcumin present) ; Beta = 0.15 (increased proliferation rate in TNBC via Hedgehog/ Glil pathway) ; Rho = 0.01 (reduced immune clearance, due to stronger immune evasion in TNBC).

Expected simulation dynamics: Tumor expands more aggressively compared to scenarios 1 & 2 (beta = 0.15). Immune clearance is reduced due to strong immune evasion. The overall tumor suppression is reduced due to the mode of administration and lower response of epigenetic mechanisms. Treated cells (yellow) persist longer (T_treat = 8), before transitioning to dead cells (grey). Residual cancer cells (red) persist at late time steps, reflecting the clinical reality of TNBC therapeutic challenges.



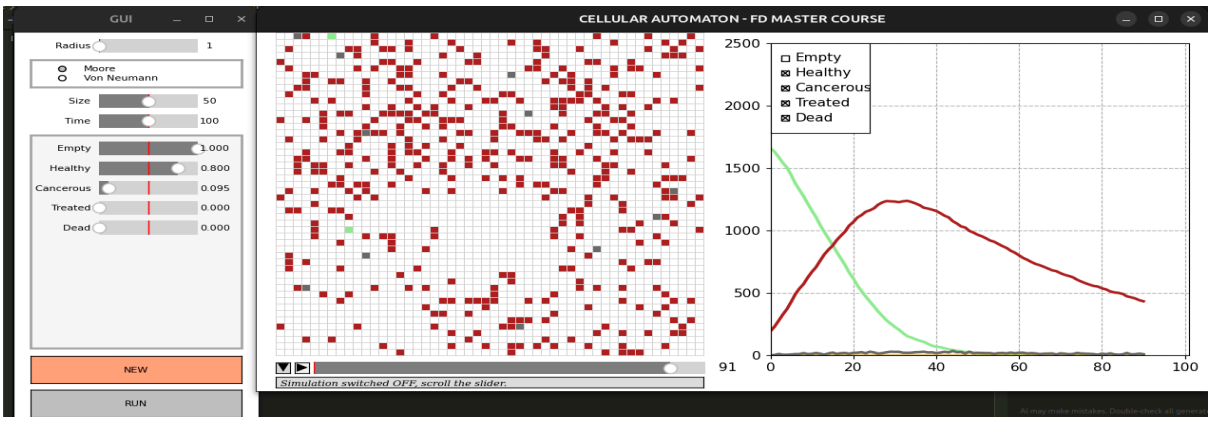
❖ Scenario 4: No treatment (Control group) (`SCENARIO_NO_TREAT = True`)

Biological question: What happens to the tumour without any curcumin intervention, driven only by spontaneous immune clearance ?

Biological justification: All three articles include a control group without curcumin treatment to define a baseline, helping to highlight and quantify curcumin's therapeutic effect. In the study "Cao et al., 2022", they use a saline/ DMSO control in the xenograft mouse model. In "Mayo et al., 2024", a placebo is used in the clinical trials. We note that without curcumin, the tumor growth is governed solely by contact driven transformation and with immune clearance, which are insufficient to inhibit the progression of tumor in these conditions.

Parameters: gamma_admin = 0 (no drug delivery) ; EPI = 0 (no epigenetic effect) ; Curcumin_present = False(no curcumin) ; Beta = 0.1 (standard proliferation rate) ; Rho = 0.02 (standard immune clearance, insufficient to stop tumor).

Expected simulation dynamics: We have unrestricted tumor expansion, where cancerous cells (red) propagate outward through healthy tissue, from direct cell-to-cell contact transformation at every step. No treated (yellow) or dead (grey) cells appear beyond a small baseline immune clearance (rho). The grid progressively fills up with cancerous (red) cells as healthy tissue (green) is displaced. This scenario provides a reference to which all treated scenarios can be compared.



❖ Administration route: (Oral, IV, Nano)

The administration route is selected via a pop-up at the launch of the program. This can affect the shape of D(t) and determines the parameter y_{gamma} in any simulation.

Parameters: Oral (y_{gamma} = 0.2, due to low bio-availability) ; IV (y_{gamma} = 0.7, better bio-availability than oral administration) ; Nano (y_{gamma} =0.85 , better bio-availability).

5. Conclusion

The current study modeled the biological effect of curcumin on breast cancer using 2D Cellular Automata. Through the use of empirical data from studies conducted in the clinical and molecular sciences, it was possible to translate biological behavior into local transition functions and demonstrate the emergence of intricate tissue structures from simple cellular interactions.

The findings revealed that curcumin is capable of stimulating several anti-tumor responses, such as ferroptosis (cell death caused by iron ions) and the activation of tumor suppressor genes via epigenetic modification.

However, the findings also indicated that the efficacy of curcumin is significantly dependent on the dosage form. As mentioned earlier, conventional oral administration with limited absorption rates could not overcome aggressive tumors like TNBC. On the other hand, the use of nanoparticles for drug delivery yielded better outcomes since the medication led to the elimination of tumors and restoration of healthy tissues.

A possible extension to the model would be the addition of an angiogenesis mechanism, allowing tumors to create new blood vessel pathways once cancer cell density becomes high enough. This would increase the biological realism and complexity of the simulation.

6. References

Mayo, B., Penroz, S., Torres, K., & Simón, L. (2024). **Curcumin Administration Routes in Breast Cancer Treatment.** *International journal of molecular sciences*, *25*(21), 11492. https://doi.org/10.3390/ijms252111492

Cao, X., Li, Y., Wang, Y., Yu, T., Zhu, C., Zhang, X., & Guan, J. (2022). **Curcumin suppresses tumorigenesis by ferroptosis in breast cancer.** *PloS one*, *17*(1), e0261370. https://doi.org/10.1371/journal.pone.0261370

Fabianowska-Majewska, K., Kaufman-Szymczyk, A., Szymanska-Kolba, A., Jakubik, J., Majewski, G., & Lubecka, K. (2021). **Curcumin from Turmeric Rhizome: A Potential Modulator of DNA Methylation Machinery in Breast Cancer Inhibition.** *Nutrients*, *13*(2), 332. https://doi.org/10.3390/nu13020332

[1] Bayet-Robert, M., Kwiatowski, F., Leheurteur, M., Gachon, F., Planchat, E., Abrial, C., Mouret-Reynier, M.-A., Durando, X., Barthomeuf, C., & Chollet, P. (2010). **Phase I dose escalation trial of docetaxel plus curcumin in patients with advanced and metastatic breast cancer.** *Cancer Biology & Therapy, 9*(1), 8–14. https://doi.org/10.4161/cbt.9.1.10392

[2] Saghatelyan, T., Tananyan, A., Janoyan, N., Tadevosyan, A., Petrosyan, H., Hovhannisyan, A., Hayrapetyan, L., Arustamyan, M., Arnhold, J., Rotmann, A.-R., et al. (2020). **Efficacy and safety of curcumin in combination with paclitaxel in patients with advanced, metastatic breast cancer: A comparative, randomized, double-blind, placebo-controlled clinical trial.** *Phytomedicine, 70*, 153218. https://doi.org/10.1016/j.phymed.2020.153218

[3] Rahman, M. A., Mittal, V., Wahab, S., Alsayari, A., Bin Muhsinah, A., & Almaghaslah, D. (2022). **Intravenous nanocarrier for improved efficacy of quercetin and curcumin against breast cancer cells: Development and comparison of single and dual drug-loaded formulations using hemolysis, cytotoxicity and cellular uptake studies.** *Membranes, 12*(7), 713. https://doi.org/10.3390/membranes12070713

[4] Norouzi, S., Majeed, M., Pirro, M., Generali, D., & Sahebkar, A. (2018). **Curcumin as an adjunct therapy and microRNA modulator in breast cancer.** *Current Pharmaceutical Design, 24*(2), 171–177.

https://doi.org/10.2174/1381612824666171129203506

[5] Luo, M., Wu, L., Zhang, K., Wang, H., Zhang, T., Gutierrez, L., et al. (2018). **miR-137 regulates ferroptosis by targeting glutamine transporter SLC1A5 in melanoma.** *Cell Death & Differentiation, 25*(8), 1457–1472. https://doi.org/10.1038/s41418-017-0053-8

[6] James, S. J., Pogribny, I. P., Pogribna, M., Miller, B. J., Jernigan, S., & Melnyk, S. (2003). **Mechanisms of DNA damage, DNA hypomethylation, and tumor progression in the folate/methyl-deficient rat model of hepatocarcinogenesis.** *The Journal of Nutrition, 133*(11 Suppl. 1), 3740S–3747S. https://doi.org/10.1093/jn/133.11.3740S

[7] Dandawate, P. R., Subramaniam, D., Jensen, R. A., & Anant, S. (2016). **Targeting cancer stem cells and signaling pathways by phytochemicals: Novel approach for breast cancer therapy.** *Seminars in Cancer Biology, 40–41*, 192–208. https://doi.org/10.1016/j.semcancer.2016.09.001

[8] Hermann, A., Gowher, H., & Jeltsch, A. (2004). **Biochemistry and biology of mammalian DNA methyltransferases.** *Cellular and Molecular Life Sciences, 61*(19–20), 2571–2587. https://doi.org/10.1007/s00018-004-4201-1

[9] Milutinovic, S., Knox, J. D., & Szyf, M. (2000). **DNA methyltransferase inhibition induces the transcription of the tumor suppressor p21(WAF1/CIP1/sdi1).** *The Journal of Biological Chemistry, 275*(9), 6353–6359. https://doi.org/10.1074/jbc.275.9.6353

7. Annexes

Table (a): Summary of clinical trials to study the efficacy of curcumin with other chemotherapy in breast cancers.

Clinical Trial	Study Design	Population	Treatment	Results
Curcumin + Docetaxel	Phase I	Patients with advanced breast cancer	Curcumin (From 500 mg/day) + Docetaxel (100 mg/m^2 IV)	Enhanced therapeutic outcomes, safe dose determined
Curcumin + Radiotherapy (RT)	Randomized, double-blind, placebo-controlled clinical trial	Women with non-inflammatory breast cancer or carcinoma in situ prescribed radiation therapy (RT)	Curcumin (6 g/day)	Reduction in severity of radiation dermatitis
Curcumin + Polyphenols in Breast Cancer Surgery	Controlled trial with two groups	26 patients with diagnosed breast cancer; and 11 patients in the control group, completed the trial	Curcumin and polyphenols (three capsules/day) from diagnosis to surgery; two capsules 2–6 h before surgery	Metabolites of curcuminoids detected in mammary tissues, suggesting anti-cancer effects with long-term exposure
Curcumin + Paclitaxel	Randomized, double-blind, placebo-controlled	Patients with advanced and metastatic breast cancer	Curcumin (300 mg) + Paclitaxel (80 mg/m^2)	Improved objective response rates and reduced side effects

Table (b): Comparison between oral and intravenous routes of curcumin administration in breast cancer treatments.

Criteria	Oral Route Administration	Intravenous Route Administration
Cost	More economical	More expensive
Adherence to treatment	Easier adherence, enhanced tolerability, and suitability for an extended duration	Less adherence, reduced tolerability, and temporary use
Improved formulations	Curcumin into self-nano emulsifying drug delivery system (CPCSNEDDS) Curcumin solution dissolved in corn oi D-alpha-tocopheryl polyethylene glycol succinate -coated bilosomes were successfully formulated and loaded with curcumin (TPGS-CUR-Bil) PLGA-Levan Nanomicelle Loaded with Curcumin 1 mL/day CUR/IR780@SMEDDS 75 mg/kg (Self-Microemulsifying Drug Delivery System) Curcumin extracted (Powdered Turmeric rhizomes (500 gm)) with 100% methano Curcumin–docetaxel co-loaded nanosuspension 40 mg/kg Curcumin capsule 500 mg Capsule of 505 mg of blend (190 mg curcumin extract, 65 mg trans-resveratrol, 125 mg flaxseed extract, and 125 mg red clover extract) Curcumin (Curcumin C3 Complex®)	PEG–albumin–curcumin Nanoparticles 10 mg/mL Curcumin loaded with an amphiphilic block copolymer (MePEG-peptide-PET-PCL) nanoparticles (Cur-P-NPs) Cur-FA-CMC/PVP GO NPs Curcumin-loaded redox-responsive mesoporous silica nanoparticles (MSN/CUR-PEI-FA) Curcumin-loaded ZnO nanoparticles (ZnO-PBA-Curcumin) 10 mg/kg PLGA-PEG nanoparticles curcumin delivery system 5 mg/kg pH/redox nanocarrier CUR@PCPP Alginate–curcumin conjugate (AA-Cur) Curcumin nanocrystals (Cur-NC) modified with Hyaluronic Acid (HA@Cur-NC) Mifepristone and curcumin encapsulated in Hyaluronic Acid-modified liposomes (CUR&RU486/HA-LIPs) Curcumin-loaded and Hyaluronic Acid-coated ZIF-8 (Cur@ZIF-8@HA) Curcumin delivered Conjugating Hyaluronic Acid (HA) on the surface of Mesoporous silica nanoparticle (MSN-HA-C) Curcumin-loaded oxidation-responsive mPEG-b-PLG (Se)-TP polymeric micelle (Cur-loaded micelles) HA-CHEMS conjugates and TPGS self-assembled into Curcumine-loaded nanoparticles (HA-CHEMS-Cur-TPGS NPs) Folate-conjugated curcumin and gemcitabine-loaded nanoparticles (FCGNPs) Methotrexate and Curcumin co-encapsulated in Poly(lactic-co-glycolic acid) (PLGA) nanoparticles (MTX-CUR-NPs) Heparin-poloxamer P403 (HP403) co-load curcuminoid (Cur) and cisplatin hydrate (CisOH) (HP403@CisOH@Cur) Co-delivery of paclitaxel and curcumin by biodegradable polymeric nanoparticles (PTX-CUR-NPs) PEG-PLGA encapsulated paclitaxel and curcumin (PC-NPs) nanoparticles Curcumin® (CUC-01) 300 mg/week
Combination with other therapies	Gemcitabine Paclitaxel Docetaxel IR780 Radiotherapy	Gemcitabine Paclitaxel Methotrexate Cisplatin Doxorubicin

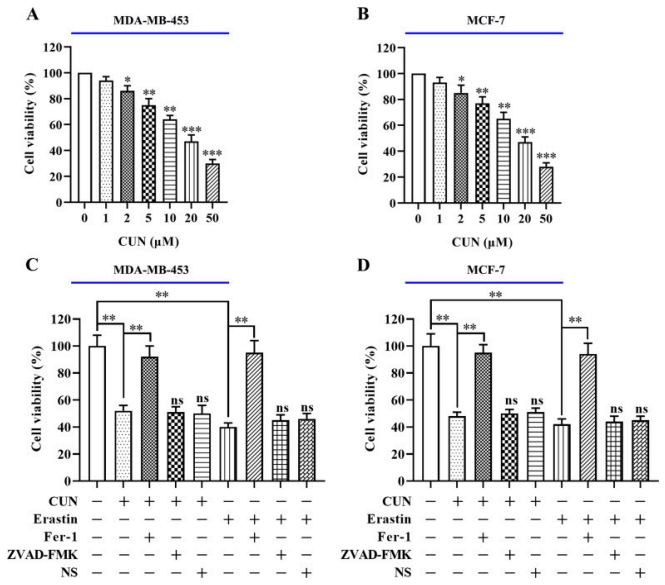


Figure (a) : Treatment with curcumin upregulates erastin-induced ferroptosis in BC cells.
*A and B: BC cells were treated with CUN at 0, 1, 2, 5, 10, 20 and 50 μM for 48 h. The CCK-8 assay was used to assess cell viability. C and D: BC cells were preincubated with various inhibitors, including ZVAD-FMK (50 μM), NS (50 μM), ferrostatin-1 (50 nM) or erastin (10 nM) for 2 h followed by CUN (20 μM) treatment for 48 h. The CCK-8 assay was performed to detect cell viability in both MDA-MB-453 and MCF-7 cells. Each data point is expressed as mean ± SEM of 3–5 independent tests. **P<0.01, ***P<0.001. CUN, curcumin; BC, breast cancer; CCK-8, cell counting kit-8; NS, necro-sulfonamide; SEM, standard error of the mean.*

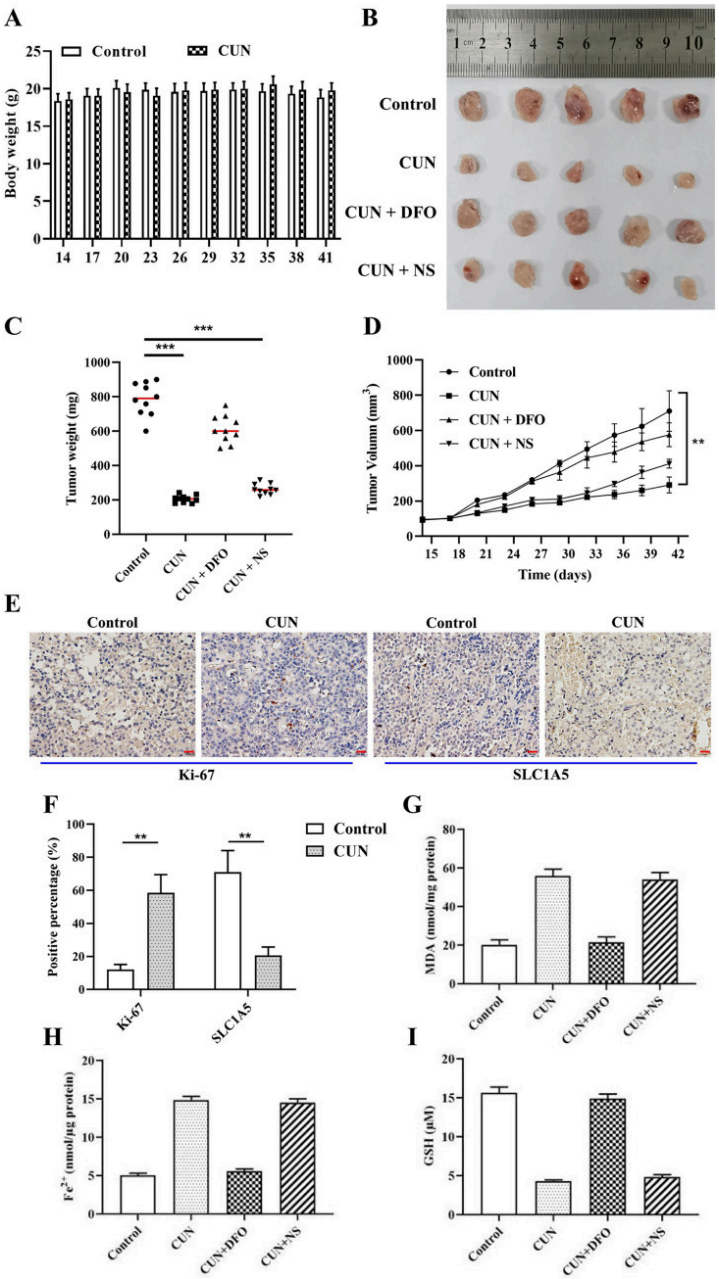


Figure b: Curcumin suppresses tumor growth in vivo.

*MCF-7 cells were subcutaneously injected into female BALB/c nude mice (n = 10/group), followed by 30 mg/kg/d curcumin treatment for 4 weeks. DFO or NS (30 mg/kg) were administered by intraperitoneal injection three times a week following administration of curcumin. A: Measurement of the body weight of mice. B: Measurement of the gross tumor size in mice; C: The tumor volume was recorded every 3 days; D: The changes in tumor weight were measured; E, F: Immunohistochemical staining of Ki-67 and SLC1A5 expressions in tumor tissues, scale bar = 20 μm; The levels of MDA accumulation (G), iron (H) and GSH (I) were evaluated in the tumor tissues. Each data point is expressed as the mean ± SEM of three independent tests. **P<0.01. CUN, curcumin; DFO, deferoxamine; NS, necro-sulfonamide; SLC1A5, solute carrier family 1 member 5; MDA, malondialdehyde; GSH, glutathione; SEM, standard error of the mean.*

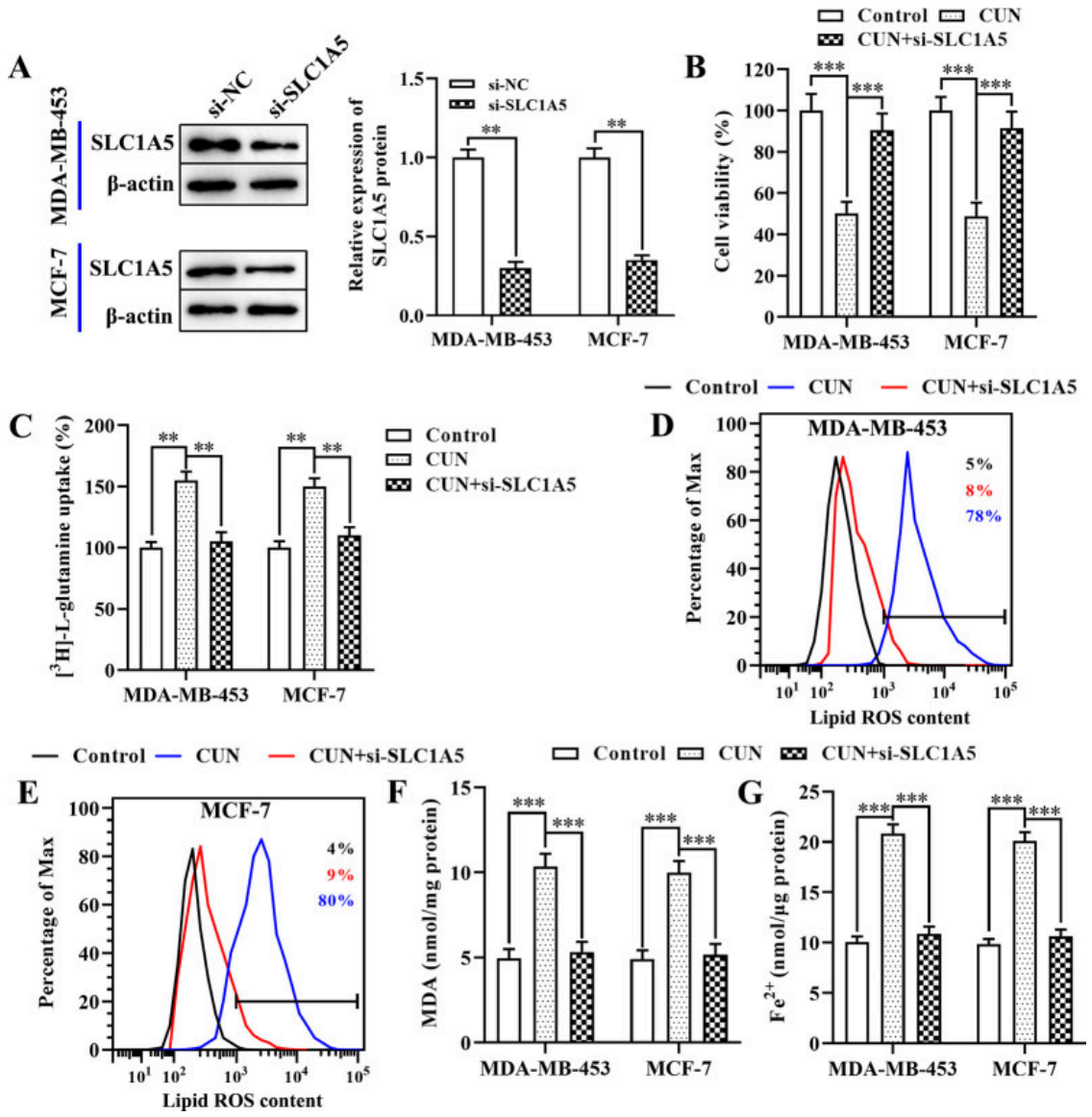


Figure c: Treatment with curcumin facilitates ferroptosis via upregulation of SLC1A5 expression in BC.

*A: Western blot analysis was performed to examine the protein levels of SLC1A5 in both BC cell lines following transfection with si-SLC1A5 or si-NC; B: The CCK-8 assay was used to examine the effects of SLC1A5 knockdown on cell viability in both BC cell lines; C: The [^{3}H]-L-glutamine uptake was determined using a liquid scintillation counter; D and E: Flow cytometry was employed using C11-BODIPY to detect the levels of lipid ROS in BC cell lines treated with curcumin or erastin; F: The MDA accumulation in both BC cells was examined by a lipid peroxidation assay kit; G: The intracellular Fe^{2+} levels in both BC cell lines were measured by an iron assay kit. Each data point is expressed as mean ± SEM of 3–5 independent tests. **P<0.01. CUN, curcumin; SLC1A5, solute carrier family 1 member 5; BC, breast cancer; si-SLC1A5, SLC1A5 small interfering RNA; si-NC, NC-small interfering RNA; CCK-8, cell counting kit-8; ROS, reactive oxygen species; MDA, malondialdehyde; SEM, standard error of the mean.*